Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	*(a)*	(i)		Boxes 2 and 3 ticked: (1) + (1) **Note** – for each extra tick subtract 1 mark. No negative mark.		2		2		
		(ii)		Kepler 11 has <u>same mass</u> as our Sun (1) Kepler 11 has [the most] <u>similar</u> temperature to our Sun (1) Each additional item of data stated – 1 (minimum 0)			2	2		
	(b)	(i)		The exoplanet blocks out / absorbs [some] light from the star Accept: there is a shadow / eclipse 'The planet doesn't give out light' – not enough	1			1		
		(ii)	I	2 orbits take 150 [days] or 300-150 [days] (1) Single orbit = $\frac{150}{2}$ = 75 [days] (1) [Accept 75 – 78 days] **Alternative** 4 orbits take 300 [days] (1) [accept 310 days] Single orbit = $\frac{300}{4}$ = 75 [days] (1) [Accept 75 – 78 days]			2	2	2	
			II	An extra/different dip in the intensity line is present [accept anomaly] Not: another transit shown			1	1		
			III	[An absorption spectrum arises because] gases absorb some light (1) Different gases [accept: elements] [in the planet's atmosphere] have <u>different</u> black lines / [absorb light at] different wavelength / frequencies . (1)	2			2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		In both systems the temperature decreases with distance / orbit radius (1) But In the Solar System Venus is the 2nd planet but the hottest / Mercury is the first and not the hottest (1) [i.e the nature of the anomaly must be made clear]			2	2		
		(ii)		0.25 – 0.26		1		1	1	
				Question 9 total	**3**	**3**	**7**	**13**	**3**	**0**